

Product Requirements Document (PRD)

Centralized Real-Time Queue Management Platform

Document Information

Field	Details
Product Name	Centralized Real-Time Queue Management Platform
Document Type	Product Requirements Document
Version	1.0
Status	Draft
Product Type	Centralized Queue Management Application
Primary Users	End Users, Queue Operators, Organization Administrators, Platform Administrators
Primary Experience	Mobile-first Web Application + Centralized Management Dashboard

1. Executive Summary

The Centralized Real-Time Queue Management Platform is a configurable platform designed to digitally manage waiting queues for different types of organizations and services. The platform is not limited to restaurants. It can be used anywhere people need to wait for a

service, resource, counter, person, appointment, or turn.

Potential use cases include:

- Restaurants
- Government offices and chambers
- Hospitals and clinics
- Banks
- Educational institutions
- Public and customer service centers
- Retail stores and transportation/service counters
- Event venues

The core platform provides a centralized mechanism for organizations to create and manage queues while allowing people to join a queue, view their position, receive real-time updates, and receive notifications when their turn is approaching or available.

Core Experience Flow:

Person -> Join Queue -> Receive Queue Position -> Real-Time Queue Updates (#4 -> #3 -> #2 -> #1) -> Your Turn -> Notification -> Proceed to Service / Counter / Resource

2. Problem Statement

Many organizations still manage queues using physical lines, paper tokens, manual calling, spreadsheets, or isolated location-based systems. These create severe friction for both parties:

- **Problems for People Waiting:** Lack of visibility on wait times, forced physical presence in waiting areas, anxiety of missing announcements or turn calls.
- **Problems for Organizations:** Excessive staff time spent managing manual queues, handling dropouts, no-shows, and manual notifications across multiple service points.

3. Product Vision & Goals

The product aims to become a single centralized platform where any organization can create and manage digital queues while keeping generic queue engines separated from organization-specific workflows.

Product Goals:

- Allow organizations to create and manage digital queues seamlessly.
- Provide a simple, mobile-first experience for end users joining queues without mandatory app downloads.
- Deliver real-time queue position updates (#4 -> #3 -> #2 -> #1) and handle sudden queue changes dynamically.
- Support multi-channel notifications (WhatsApp, SMS, Email, etc.) when turns open up.
- Provide scalable security, multi-organization isolation, rate limiting, and comprehensive testing (happy & negative path).

4. Product Principles

Principle	Description
Generic by Design	Core platform concepts remain identical ("YOUR TURN IS AVAILABLE") regardless of industry terminology (Table, Counter, Room, Desk).
Configurable	Supports organization-specific queue rules, registration fields, notification behavior, and custom branding without hardcoded overrides.
Real-Time First	Queue states reflect changes instantaneously without needing manual browser refreshes.
Production Ready	Every feature must be secured, validated, covered by automated tests, and fully documented before being marked DONE.

5. Target Users & Roles

- **End User / Person Waiting:** Customer, Patient, Citizen, Student, or Applicant seeking service.
- **Queue Operator:** Staff member (receptionist, bank teller, desk officer) calling next candidates, updating statuses, or managing cancellations/no-shows.
- **Organization Administrator:** Manages settings, locations, services, operators, resources, rules, and notification behavior for their specific entity.
- **Platform Administrator:** Super-admin handling system-wide health, multi-tenant organizations, and overall platform configurations.

6. Supported Industry Use Cases

Industry	Service Unit	Turn Notification Message
Restaurant	Dining Queue / Tables	"Your table is ready."
Government Office	Service Window / Application Desk	"Please proceed to Counter 3."
Hospital / Clinic	Doctor / Department Consultation	"Please proceed to Room 204."
Bank	Customer Service / Teller Counter	"Please proceed to Counter 5."

7. Queue Entry & Registration

Queue joining can occur via multiple touchpoints, including **QR Codes, Web Links, Kiosks, Direct Websites, or Staff Manual Entry.**

Registration forms are dynamically generated based on the organization's configuration:

- **Restaurant:** Name, Phone, Party Size, Seating Preference.
- **Government:** Name, Phone, Service Type, Application Number.

- **Hospital:** Name, Phone, Department, Patient ID/Info.

8. Queue Progression & Turn Mechanics

- **Position Assignment & Real-Time Sync:** Assigns positions based on rules (First-come-first-served, Priority, VIP) and updates in real-time.
- **Sudden Changes & Queue Clearing:** If entries clear out unexpectedly, downstream positions immediately re-calculate and trigger turn notices if necessary.
- **Operator Call Action:** Operator clicks "Call Next" (or custom equivalent), triggering real-time state changes to CALLED and sending external notifications.
- **Claim / Response Window:** Configurable timeout windows (e.g., 10 minutes) for users to acknowledge their turn before being marked EXPIRED or NO_SHOW.

9. Queue State Lifecycle

The generic state engine supports the following transitions:

```

WAITING ---> CALLED ---> RESPONDED ---> IN_SERVICE ---> COMPLETED
|
+----> EXPIRED / NO_SHOW

```

10. Security, Isolation & Compliance

- **Environment Security:** Secrets, database credentials, and notification API keys strictly managed via uncommitted .env files.
- **Authentication & Authorization:** Role-based access control (RBAC) ensuring Operators and Admins can only view/manage their assigned tenant queues.
- **API Protection & Rate Limiting:** Strict rate limits on sensitive/public APIs to prevent spam or service disruption.
- **Data Privacy:** Minimally required collection of personal data; phone numbers and credentials are protected and never publicly exposed.

11. Testing & Code Quality Framework

Testing Strategy	Tooling & Details
Unit Testing	Core queue logic, state transitions, validation, and notification event generation.
End-to-End (E2E)	Automated using Playwright (and Selenium where required) for complete user journeys.
Happy & Negative Paths	Covers full success workflows as well as invalid registration, reconnects, expired turn calls, and unauthorized requests.
Quality Checks	Automated pre-commit hooks via Husky running linters, type checks, and formatters.

12. Proposed Roadmap & Phased Rollout

- **Phase 1 — Core Centralized Queue:** Organization setup, Queue creation, Mobile joining, Real-time status sync, Operator Dashboard.
- **Phase 2 — Turn & Notification Engine:** Operator call mechanics, WhatsApp/SMS integration, User turn responses, Completion flows.
- **Phase 3 — Production Hardening:** Auth, RBAC, Rate limiting, Backend validation, Playwright E2E tests, Observability.
- **Phase 4 — Multi-Organization Configuration:** Tenant management, custom fields, configurable notification triggers, custom branding.
- **Phase 5 — Advanced Analytics & Capabilities:** Priority/VIP queues, resource allocation, wait time forecasting, system integrations.